Case Study 2: Golden State Warriors

By

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“On my honor as an Aggie, I have neither given nor received unauthorized aid on this academic work.”

This is entirely my own work or—if authorized by my instructor—the work of the team I was assigned to, except as documented in a bibliography or acknowledged below.

Signature * /*Nikitha Joshy*/

Artificial Intelligence (AI) Acknowledgement:

Google Gemini was used to help create the plots and figures used for analysis and to create the models, and for rephrasing some sections in the Methods portion of the report.

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Executive Summary

This project was undertaken on behalf of the Golden State Warriors (GSW) to create ML models that could estimate how much in advance someone would buy a game ticket and classify someone as a last-minute, early, or in-between purchaser of tickets. For the task, both a Random Forest Regressor and a Gradient Boosting Model were created, and both outperformed the Linear Regression model and the Decision Tree created by the GSW. With the model that classifies customer types, GSW could see an increase in about $40,000 of revenue by just getting 10% more customers who buy tickets more than 2 weeks in advance. So, by promoting tickets to older fans on the mailing list, GSW can reach the fans that spend more on tickets and on concessions. About 75% of fans that buy their tickets early also tend to prefer being seated in the lower deck, and lower deck tickets on average got the GSW about $590 more per customer than revenue from customers who chose upper deck seats. The fan mailing list was also a contributor to revenue, with fans on the mailing list spending on average about $298 more than those not on the list. A Facebook marketing campaign to reach this early ticket buying target audience could be vital to a massive increase in revenue. GSW should also keep up their data collection to see how Covid, shifting from the Oracle to Chase Stadium, and team success affects their ticket sales and game turnout.

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1. Introduction

1.1 Introduction to the Golden State Warriors

The Golden State Warriors (GSW) are a professional basketball team who play in the NBA (National Basketball Association). Raye Yocke, the Senior Director of Business Analytics and Strategy for the GSW, was reflecting on how much they have grown in the last couple years, and an ML model was created. The model, called "The Ticket Timing Model," changed the organization's approach to selling tickets. Typically, GSW would send mass emails about unsold tickets to anyone that was on their mailing list. However, Yocke believed an ML model could be created to predict whenever an individual fan would be most likely to buy a ticket. From market research, 3 types of customers were identified: las minute purchasers, planners (by tickets more than 2 weeks in advance), and people who purchase between these 2 timeframes.

1.2 Project Goals and Objectives

For Case Study 2, there were a couple main objectives. The two main objectives were to identify a better predictive model for identifying which category a given customer belonged to, and if a better model could be generated to estimate the number of days before a game a customer is likely to purchase a ticket. So, new models will need to be created based on the analysis Yocke's team already has done and then those will be compared to Yocke's to see which is better.

2. Methods

2.1 Estimating how Early Customers Buy Tickets

To estimate how early customers buy tickets, the GSW used simple and multiple linear regression, however in this case study, a Random Forest Regressor (RFR) was chosen to create the model. A Random Forest works like an ensemble learning method, making many individual decision trees during the training phase. For a given customer, each decision tree in the forest generates predictions for the target variable. The model's final output is then determined by averaging the predictions from all trees and yields a model that is highly robust and resistant to overfitting. An RFR was chosen because it can capture nonlinearities and any interactions between data automatically and can also train in parallel so it is fast, which will be convenient for GSW to employ. In this case, the target variable was 'Days_Before_Game', and the features used for training were customer age, whether they were on the fan mailing list, the number of tickets they purchased, seat location, ticket price, and they amount they spent on concessions. After splitting the train and test data, the model was fit and evaluated.

2.2 Classification of Customer types

The approach GSW took to classify customers into 3 types: planners, last-minute purchasers, and in between, was to use a decision tree model. GSW also tested their model on a small dataset, and they got some interesting results. For this study, to classify customer type, a gradient boosting model was created. The model fits an initial tree to the data, identifies the customers it misclassified, and then trains the next tree to specifically address those errors. This process allows the model to improve over time by focusing on the most challenging examples. So, it becomes very good at learning intricate relationships within the data. Furthermore, Gradient Boosting includes built-in regularization, a key feature that prevents overfitting by controlling model complexity. This ensures that the model can generalize to new customers. In this case, it was important to be able to identify relationships between different variables to be able to classify a customer as a certain category, so gradient boosting is a good choice, and it also outperforms decision trees because it builds many trees sequentially.

3. Results

Table 1 below shows the results of GSW's simple and multiple regression models predicting how many days before game a customer would purchase a ticket.

Table 1. GSW's Regression Results

customerID	Days Before Game	Simple	Multiple
1	28	16.3	27.0
2	1	5.0	1.9
3	3	15.1	9.0
4	7	14.0	12.7
5	21	14.0	15.5

Table 2 below displays the random forest regressor predictions on the same data for the days before a game a customer would purchase a ticket.

Table 2. RFR Prediction Results

customerID	Days Before Game	RFR Results
1	28	23.4
2	1	3.0
3	3	7.2
4	7	7.0
5	21	20.6

Figure 1 below displays the Decision Tree generated by the GSW team to classify customer types.

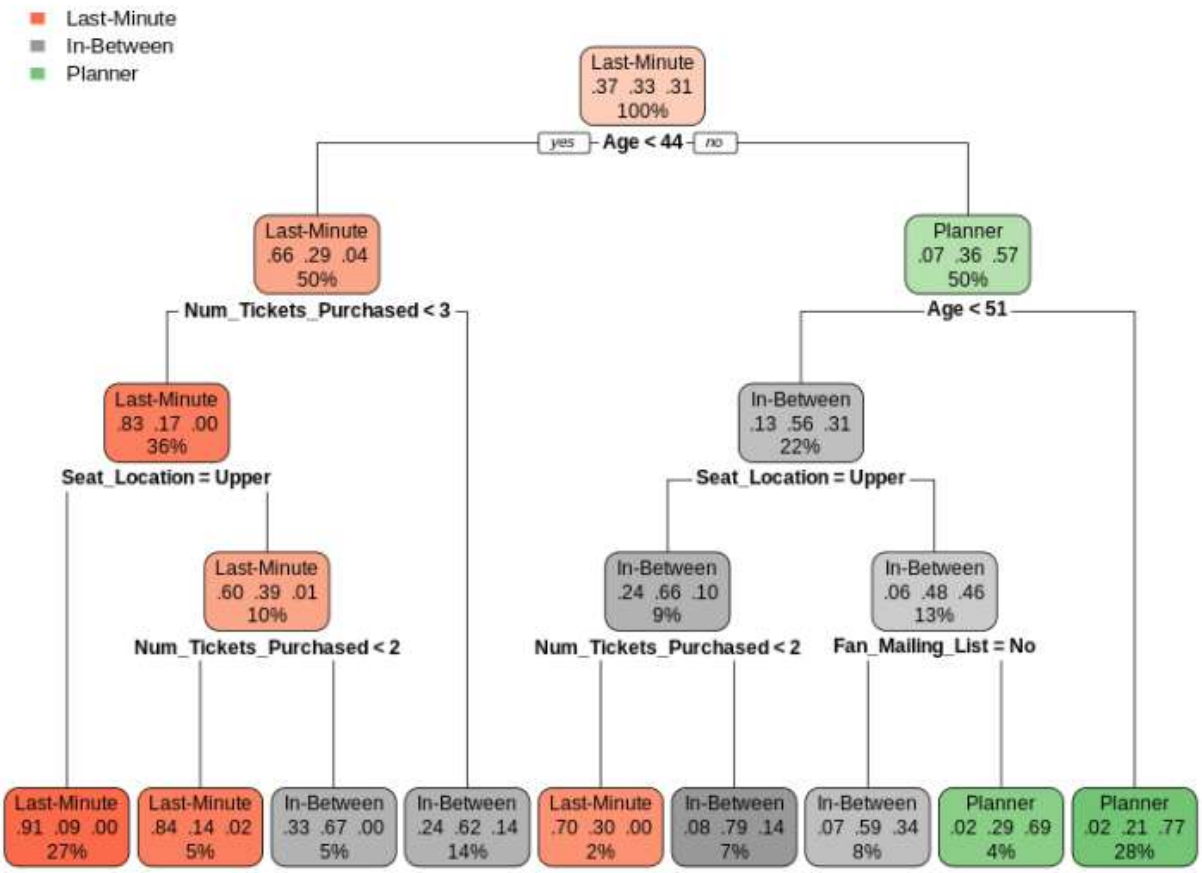


Figure 1. Decision Tree for Classifying Customer Types by the GSW

Table 3 below displays the probability of the decision tree classifying different customers.

Table 3. Probabilities of the Decision Tree Classifying Customer Types

customerID	Customer Type	Last-minute probability	In-between probability	Planner probability
1	Planner	0.02	0.21	0.77
2	Last-minute	0.91	0.09	0.00
3	Last-minute	0.70	0.30	0.00
4	In-between	0.07	0.59	0.34
5	Planner	0.07	0.59	0.34

Table 4 below shows metrics for the Gradient Boosting Model classifying customer types.

Table 4. Metrics for Gradient Boosting Model classifying Customer Types

Customer type	Precision	Recall	F1-score
Last-Minute	0.93	0.88	0.90
In-Between	0.71	0.74	0.72
Planner	0.81	0.82	0.81

Table 5 displays the revenue per customer type based on ticket and concession purchases.

Table 5. Ticket and Concession Revenue by Customer Type

Customer type	Ticket Revenue ($)	Concession Revenue ($)
Last-Minute	146685.06	1223.52
In-Between	294362.65	4725.53
Planner	403195.31	7570.61

Figure 2 below displays a visual representation of **Table 5**.

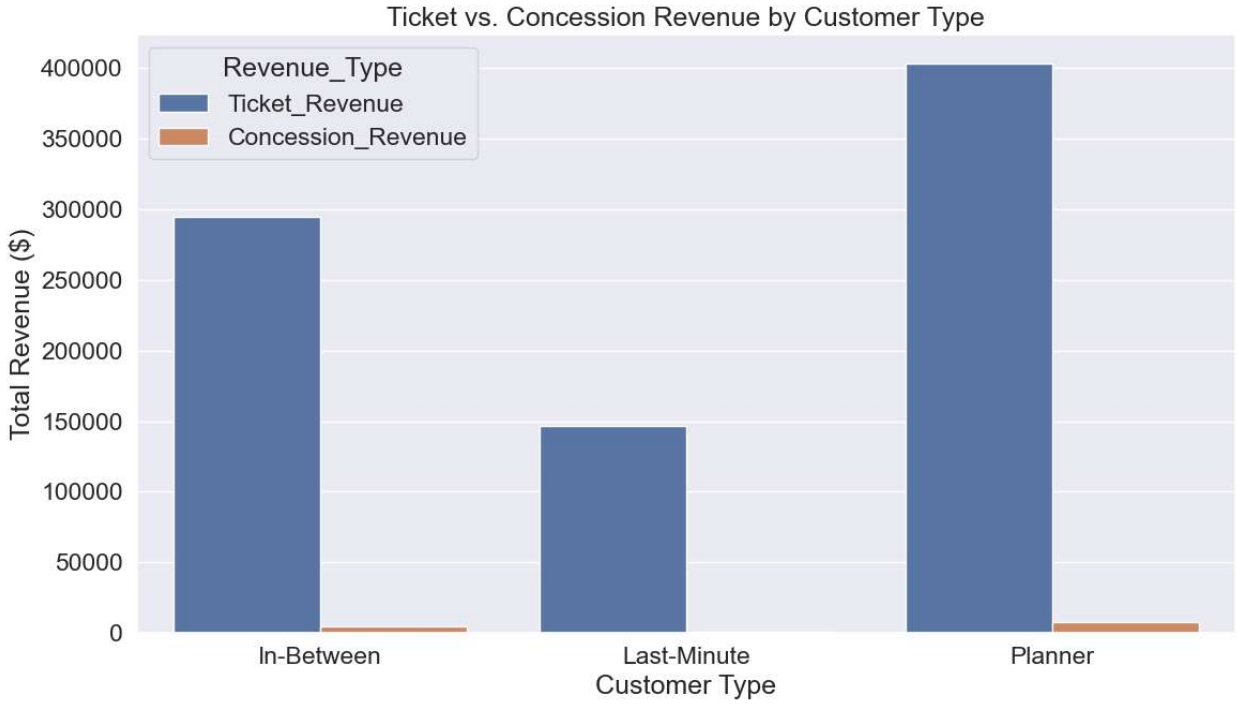


Figure 2. Visual Representation of Ticket and Concession Revenue by Customer Type

Table 6 below represents seating preference as a percentage by customer type.

Table 6.Deck Preference Based on Customer Type

Customer type	Lower Deck (%)	Upper Deck (%)
Last-Minute	19.57	80.43
In-Between	53.99	46.01
Planner	75.16	24.84

Figure 3 here represents how much people spend on average depending on whether they are on the fan mailing list.

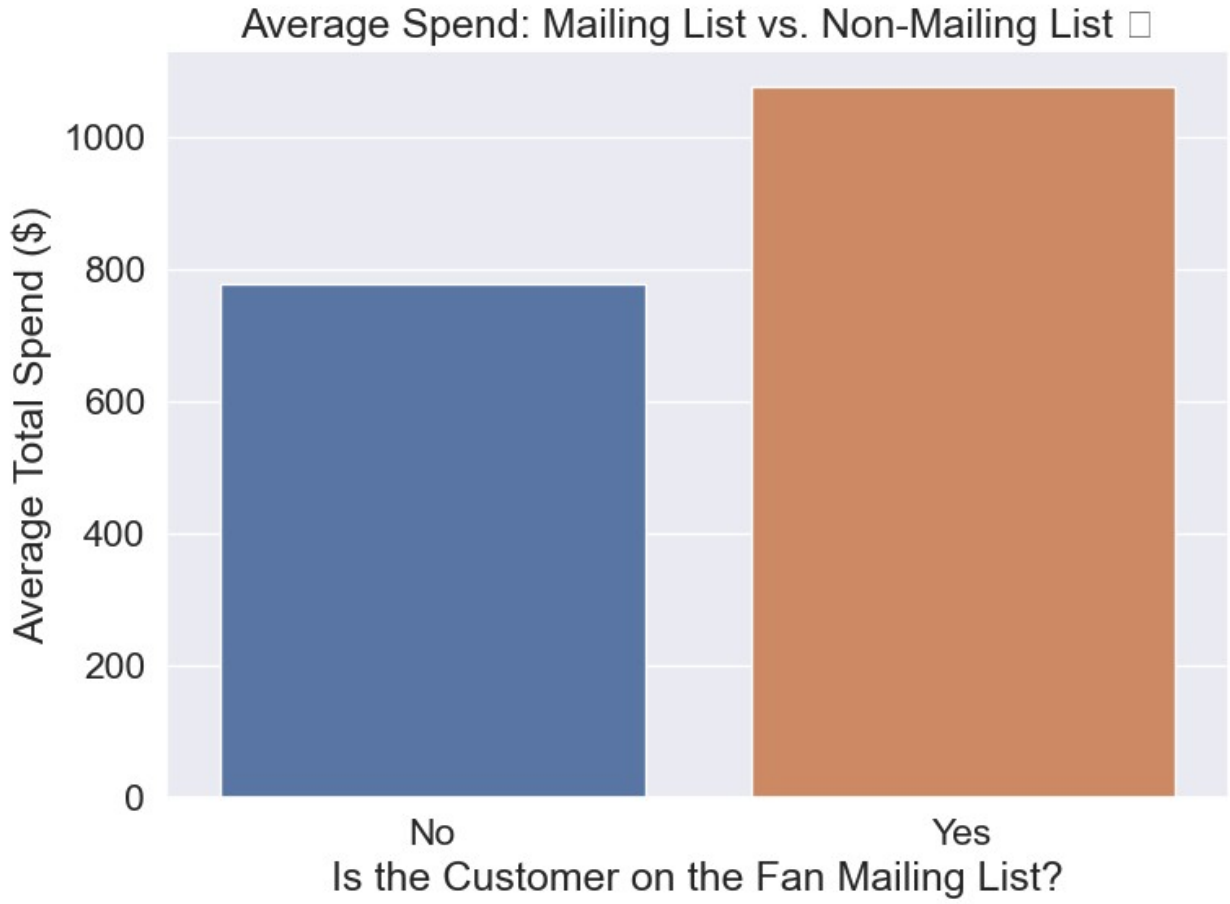


Figure 3. Average Spending based on whether a customer is on the Fan Mailing List

Figure 4 below represents how much a customer spends on both tickets and concessions based on where they are seated.

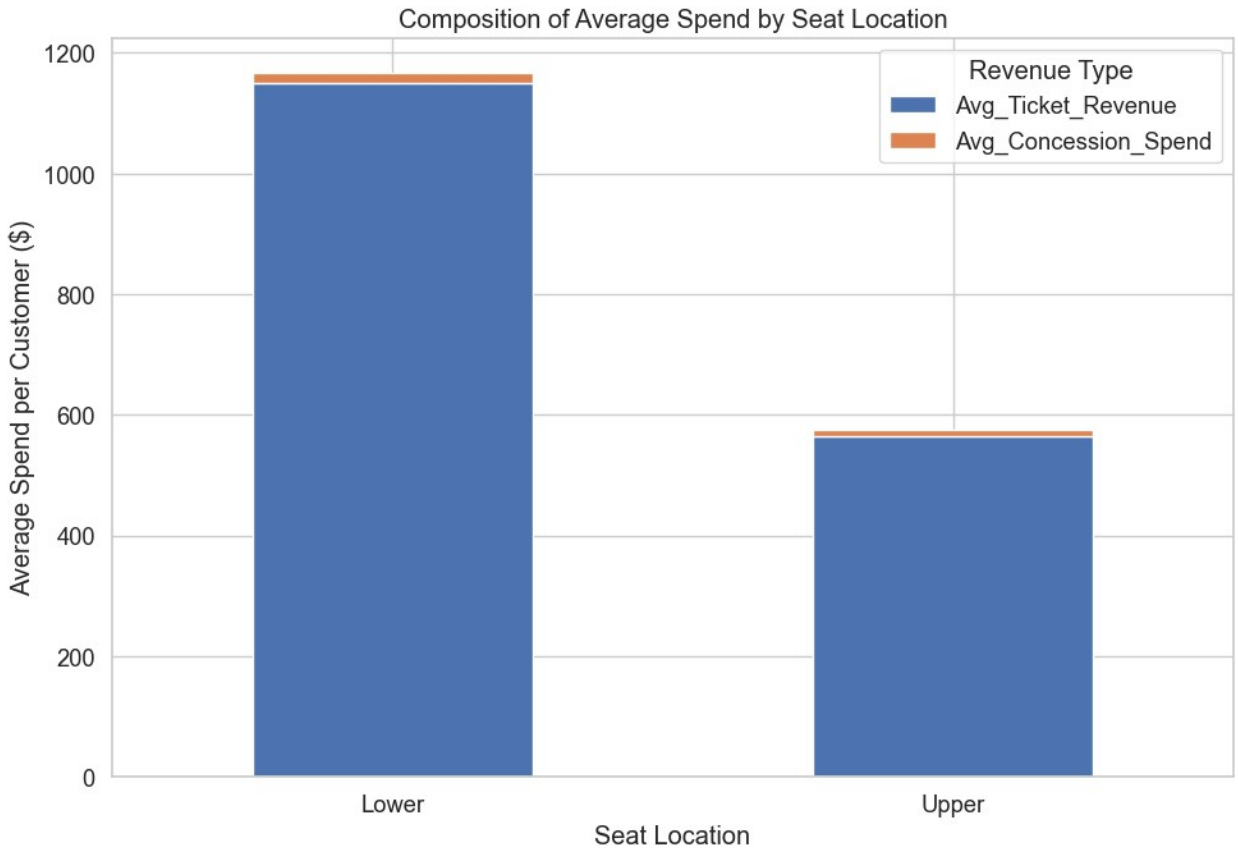


Figure 4. Average Spending per Customer based on Deck

4. Conclusions

4.1 Success of Random Forest Regressor and Gradient Boosting Model Compared to the GSW's models.

In conclusion, the random forest regressor did much better than the linear regression models in predicting how early someone would buy tickets. Using **Tables 1** and **2**, the RFR was off by around 3-4 days on estimations, but the GSW linear regression model was off by up to 10 days for customer one. So, if the GSW employed an RFR model to estimate how many days early a customer would buy tickets, they would be able to target their mailing lists towards the right group of people. For classifying customer types, the GSW had used a decision tree, but this case study employed a gradient boosting model. The GSW model was pretty good at classifying the customers based on ticket price, number of tickets bought, seat location, age, and whether a customer was on the fan mailing list, as seen in **Table 3** based off **Figure 1**. The GSW would classify as a customer as having a 0.77 chance, they are a planner when they were a planner, and a 0.91 chance they were a last-minute customer if they were a last-minute customer. From **Table 4**, the gradient boosting model created for this case study also had good success. The model was

able to get a precision of 0.93, 0.71, and 0.81 for last-minute, in-between, and planner customers respectively. So, the model could correctly predict last-minute customers 93% of the time, and it caught 88% of all last-minute customers since that was the recall rate.

4.2 Insights into Customers and Revenue

Figure 2 visualizes the total concession and ticket revenue generated by each type of customer. So, customers who bought tickets early contributed the most to ticket revenue, and to concession revenue as well. So, being able to target these planner customers would be crucial in increasing revenue for the GSW. If the number of planner customers increased by just 10%, then the total revenue would increase by more than $40,000, which is a significant amount. **Table 5** displays the exact values from **Figure 2** where the drastic difference between the revenue from last-minute customers and planners can be quantified as more than $256,000.

Certain customers also tend to buy seats in certain decks, as seen in **Table 6**, where about 75% of planners buy seats in the lower deck which is closer to the court and more expensive as **Figure 4** illustrates. So once again, planners are contributing the most the revenue from ticket sales. Only about 19% of last-minute buyers choose seats in the lower deck, while the other 81% are seated in the upper decks which are cheaper to purchase.

Finally, **Figure 3** shows how much on average a customer spends at a game (both on tickets and concessions) based on whether they are on the mailing list. Fans on the mailing list contributed on average $1076 while fans not on the mailing list contributed $778. So, the customers on the mailing list tend to increase the revenue more than those who are not. Then it would be wise to acquire more planner type customers and add them to the mailing list to ensure revenue increases.

4.3 Addressing the GSW's Concerns

The data would be representative of the future seasons when the team might not be as successful because the models use variables such as customer age, whether they are on the mailing list, seat location to estimate how much in advance they buy tickets and their customer status. So, if the GSW continue to collect data, they can observe new trends in seasons with less wins. Customer purchasing habits in the new stadium can be trained on the random forest regressor and the gradient boosting model to improve the models after the change in circumstance to ensure the most accurate results possible. So GSW would just need to keep collecting data for their centralized repository. To address the concern about post-pandemic sales, the GSW would need to perhaps send out more online promotional mail to those on their list encouraging them to buy tickets even after Covid's debilitating effects on big sports games. Then, the data of how much revenue is generated because of the mailing list can be analyzed to determine if it's still working as successfully as it used to.

5. Limitations

During the analysis, one shortcoming that seemed prominent was that with the given data, it was difficult to determine when exactly the GSW transitioned from the Oracle to Chase stadium, and whether that did affect their revenue generation. It was also not clearly stated how much of the ticket and concession sales go to the GSW, since a portion is for the stadium they play in. The GSW also took a long time to centralize their data. If this data had been collected in an organized manner for a longer period, perhaps a deeper analysis could have been conducted.

6. Recommendations

For further action by the GSW, one important recommendation is to focus on planner type customers more, especially since they generate such a large portion of both ticket and concession revenue. With the gradient boosting model, they can be identified more accurately, and the marketing team can invest more into sending promotional mail to the right customer type. Also, since it was shown that last-minute buyers purchase cheaper seats and also buy less concessions, less time and effort should be invested in that customer type. Targeting older people, because they are more likely to be in the planner category, would be crucial in increasing revenue: more than $40,000 for just a 10% increase in customers. So perhaps posting advertisements on social media platforms like Facebook could help target the right audience for ticket sales.

To address the concerns mentioned earlier, the GSW should also start tracking the dates for each game as a separate column, to see how much Covid negatively impacted them, and just going forward it would be a crucial part of creating time-series plots. Then, it would be much simpler to visualize revenue per game and per season, since the GSW might not do well every season. For marketing purposes, it would also be good to record customer gender, to be able to send out even more targeted promotional emails, and post social media campaigns aimed towards men or women. A heightened social media presence and more targeted promotional mail are crucial in driving revenue and ticket sales for the GSW.

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